

CYBERSECURITY LAB

Wazuh

SOC Homelab 2

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LAB PLATFORM	Mac host with Ubuntu and Windows virtual machines
NETWORK MODE	Shared NAT

Project Overview

The lab is operational. Wazuh 4.14.8 collects Windows Security events and detects controlled file changes on Windows and Ubuntu. Two endpoint agents are active, and their alerts are searchable in the dashboard.

What Wazuh is

Wazuh is a free, open source security platform. Its SIEM capabilities collect and analyze security events in one place. Its XDR capabilities support broader endpoint detection and response. This lab uses event collection, detection rules, file integrity monitoring, and manual investigation. [1, 2]

An agent runs inside each monitored VM. The Wazuh manager analyzes its data, the indexer stores searchable records, and the dashboard presents them in a browser. The dedicated Ubuntu 24.04 server hosts all three central components.

How this extends the first lab

The first SOC Homelab investigated packets with Wireshark and correlated Windows events with network activity. This project adds continuous endpoint monitoring. It follows the same sequence: record the observation, explain its meaning, and state only the finding supported by the evidence.

Windows generated an authentic failed-login event during a controlled test. Wazuh received that same event and produced an alert. Changes to harmless baseline files also produced file integrity alerts on both endpoints.

Verified outcomes

Validation	Result	Evidence
Central services and dashboard	Operational	WZ02
Windows and Ubuntu agents	Both Active, version 4.14.8	WZ03
Windows Security event 4625	Rule 60122, level 5	WZ04
Windows and Ubuntu file changes	Rule 550, level 7	WZ05
Agent network transport	Captured TCP port 1514 traffic	WZ06

The events were planned lab activity. Their detection validates the tested collection paths; it does not establish a compromise or prove complete coverage of every log source.

Lab Architecture and Baseline

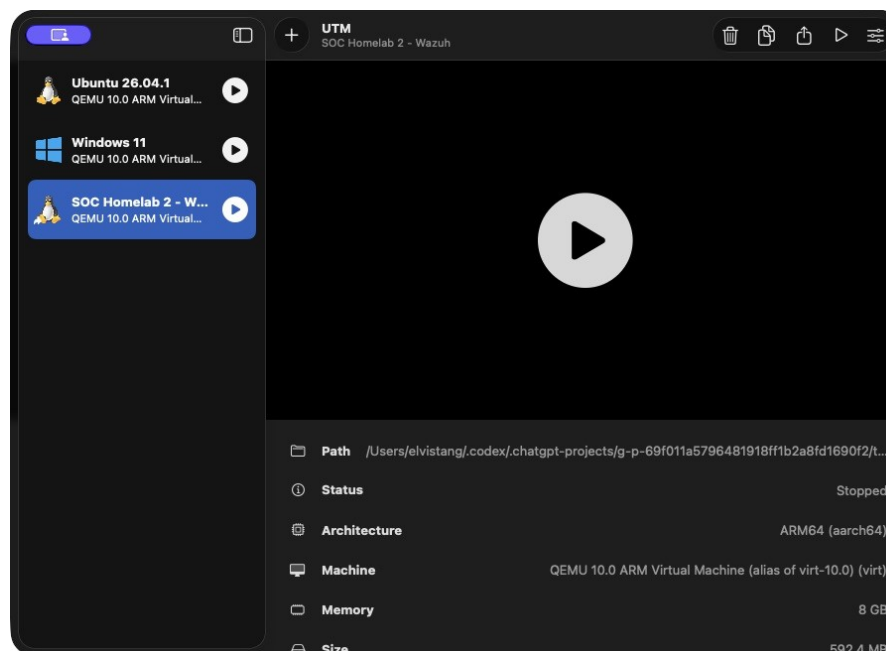
The Mac runs three ARM64 VMs in UTM on Shared NAT. Wazuh is the central server; Windows and the existing Ubuntu VM are monitored endpoints. The Mac itself has no Wazuh agent installed for this project.

System	OS and address	Allocated memory
Mac host	Apple silicon, 16 GB physical memory	16 GB total
Wazuh central server	Ubuntu 24.04.5 LTS, 192.168.64.5	8 GB; 4 vCPU; 60 GB disk
Windows endpoint	Windows 11 Pro, 192.168.64.4	4 GB
Ubuntu endpoint	Ubuntu 26.04.1 LTS, 192.168.64.3	2 GB; reduced from 8 GB

The central VM uses an official Ubuntu ARM64 cloud image. Its SHA256 matched the published checksum before deployment. Ubuntu 24.04 is listed in the Wazuh central deployment requirements. [1, 6]

Shared NAT permits the lab VMs to communicate and access installation sources. No public port forwarding was added. Allocated guest memory totals 14 GB, so close unnecessary applications and run VMs selectively if the Mac becomes slow.

Recovery snapshots were saved for the original Ubuntu and Windows disks before their agent installations. Agent configuration files were backed up before adding the test directories.

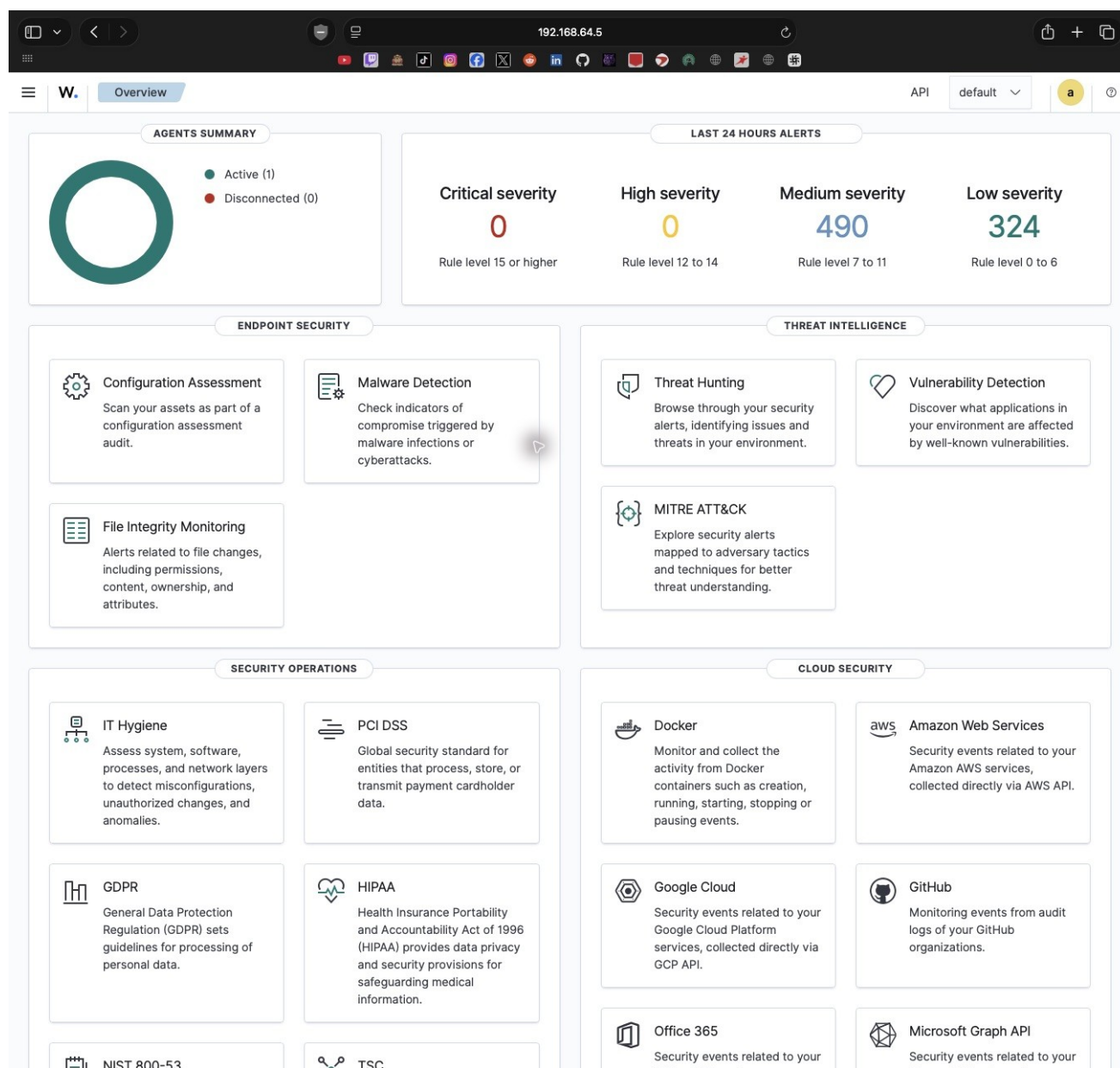


WZ01 UTM shows the dedicated Wazuh VM. This was captured during server setup; final memory and addresses are recorded above.

Central Wazuh Deployment

Wazuh 4.14.8 was installed on the dedicated Ubuntu 24.04 server using the official all-in-one installer. The manager, indexer, dashboard, and Filebeat services were all verified active. The dashboard login succeeded in Safari.

Dashboard: <https://192.168.64.5> | Server host name: wazuh-lab

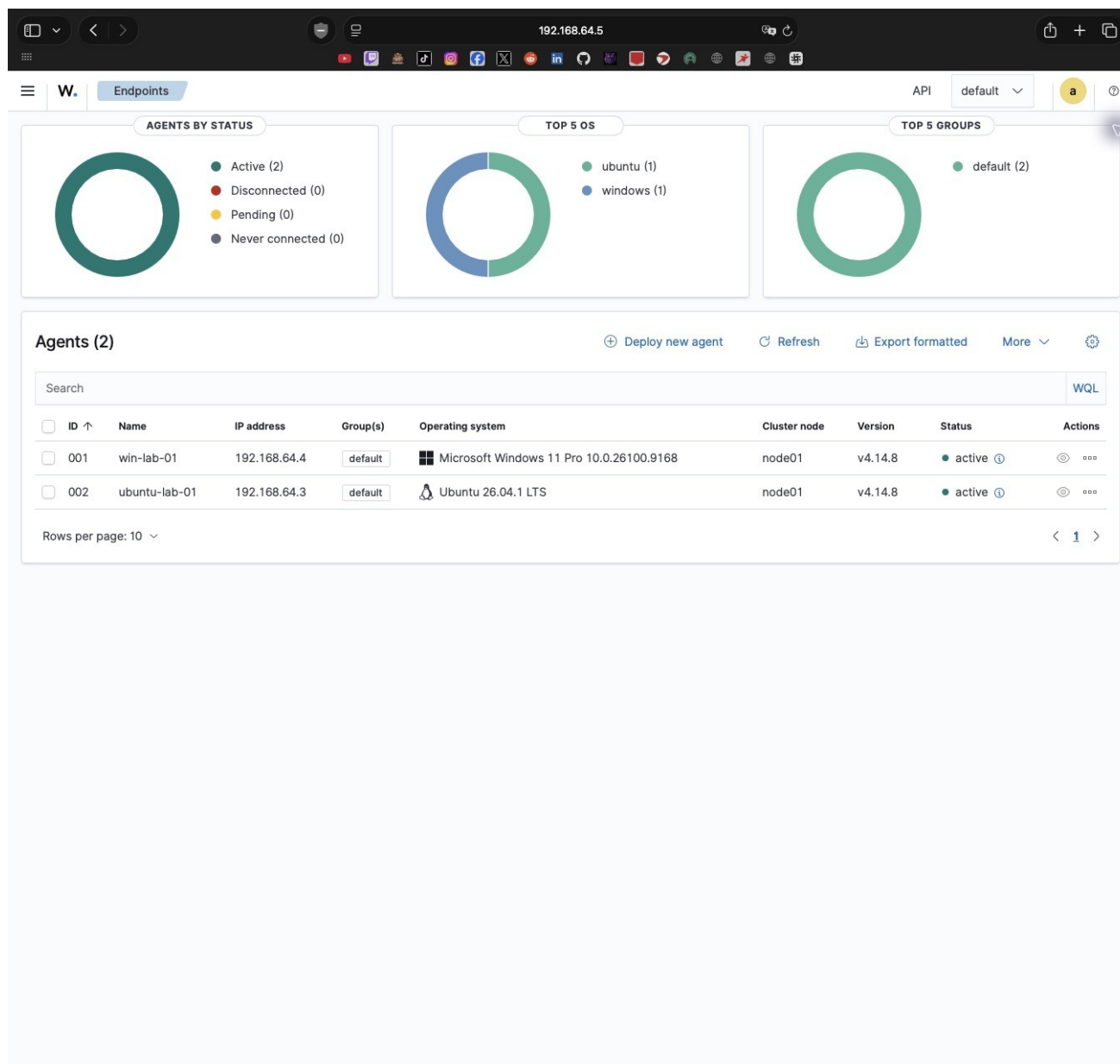


WZ02 Real dashboard after installation and Windows enrollment. Counts include startup and assessment alerts; they are not a count of attacks.

Installed packages: wazuh-manager, wazuh-indexer, and wazuh-dashboard 4.14.8-1; Filebeat 7.10.2-2. Service and version evidence is preserved in WZ02-server-status.txt.

Endpoint Enrollment and Collection

Windows is win-lab-01, agent 001. Ubuntu is ubuntu-lab-01, agent 002. Both run Wazuh 4.14.8 and were verified Active in the dashboard. Agent health was checked separately from event detection.



WZ03 Both endpoint agents are Active with their current IP addresses and installed versions.

Windows Security event-channel collection was validated with record 28868. Ubuntu collection was validated with its own file-change alert. Active status alone would not prove those sources work.

Controlled Authentication Investigation

Case ID: SOC2 AUTH 001 | Analyst: Elvis Tang | Disposition: Expected lab activity

Question and test method

Can a Windows failed-login event be matched to a Wazuh alert? A single local LogonUser API call attempted authentication as the nonexistent user SocLabDoesNotExist. The call returned failure code 1326. No real account password was changed, and this was not a successful login.

Observed evidence

Field	Matched value
Endpoint	WIN-3HRELJRC5TH / win-lab-01 / agent 001
Local source	Security event 4625, record 28 868
Account and logon type	SocLabDoesNotExist, type 2
Status and substatus	0xc000 006d / 0xc0 000 064
Wazuh alert	Rule 60 122, level 5
Rule description	Logon Failure - Unknown user or bad password

Timeline in UTC

Time on 1 October 2026	Observation
07:16:40.490	Controlled authentication test began.
07:16:40.534	Windows wrote event 4625, record 28 868.
07:16:51.667	Wazuh generated the matched alert.

The source event preceded the Wazuh alert by approximately 11.13 seconds. Dashboard display times use America/Los_Angeles, seven hours behind UTC during this test.

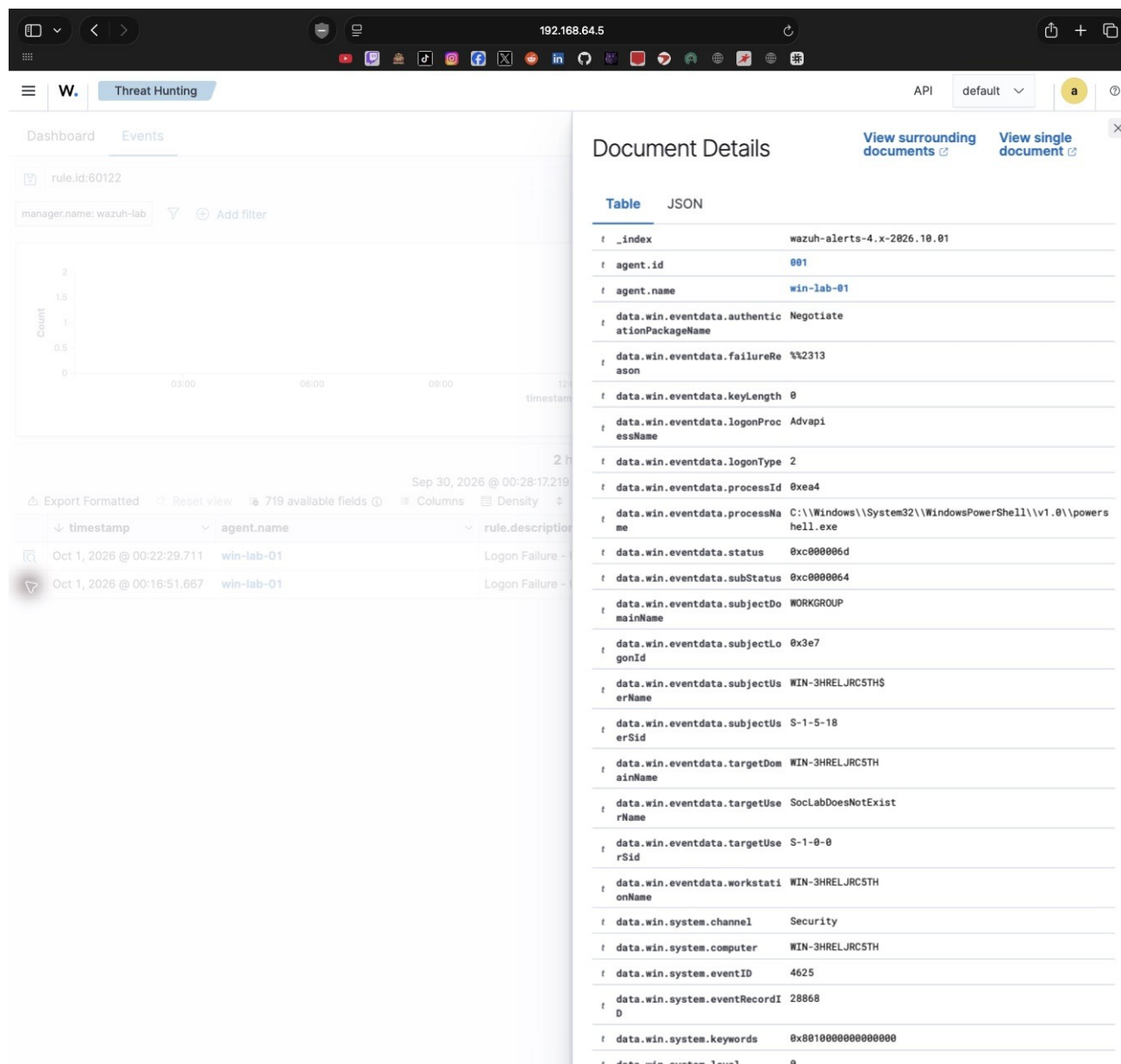
Interpretation and decision

The matching host, target user, event ID, record ID, and source timestamp establish that Wazuh received the intended Windows event. The substatus identifies the nonexistent username. The event has no source network address, so it does not support a claim of a remote attacker.

Disposition: expected lab activity, with high confidence for this matched event. No containment was needed. The rule's ATT&CK mapping is rule metadata; it does not independently demonstrate an adversary technique. Follow-up: retain the evidence and use a supported test account for a future successful-login correlation exercise.

Authentication Screenshot Evidence

The dashboard record below matches the Windows source event. The preserved source export is WZ04-windows-source.json; the matching alert is in WZ05-endpoint-alerts.json.



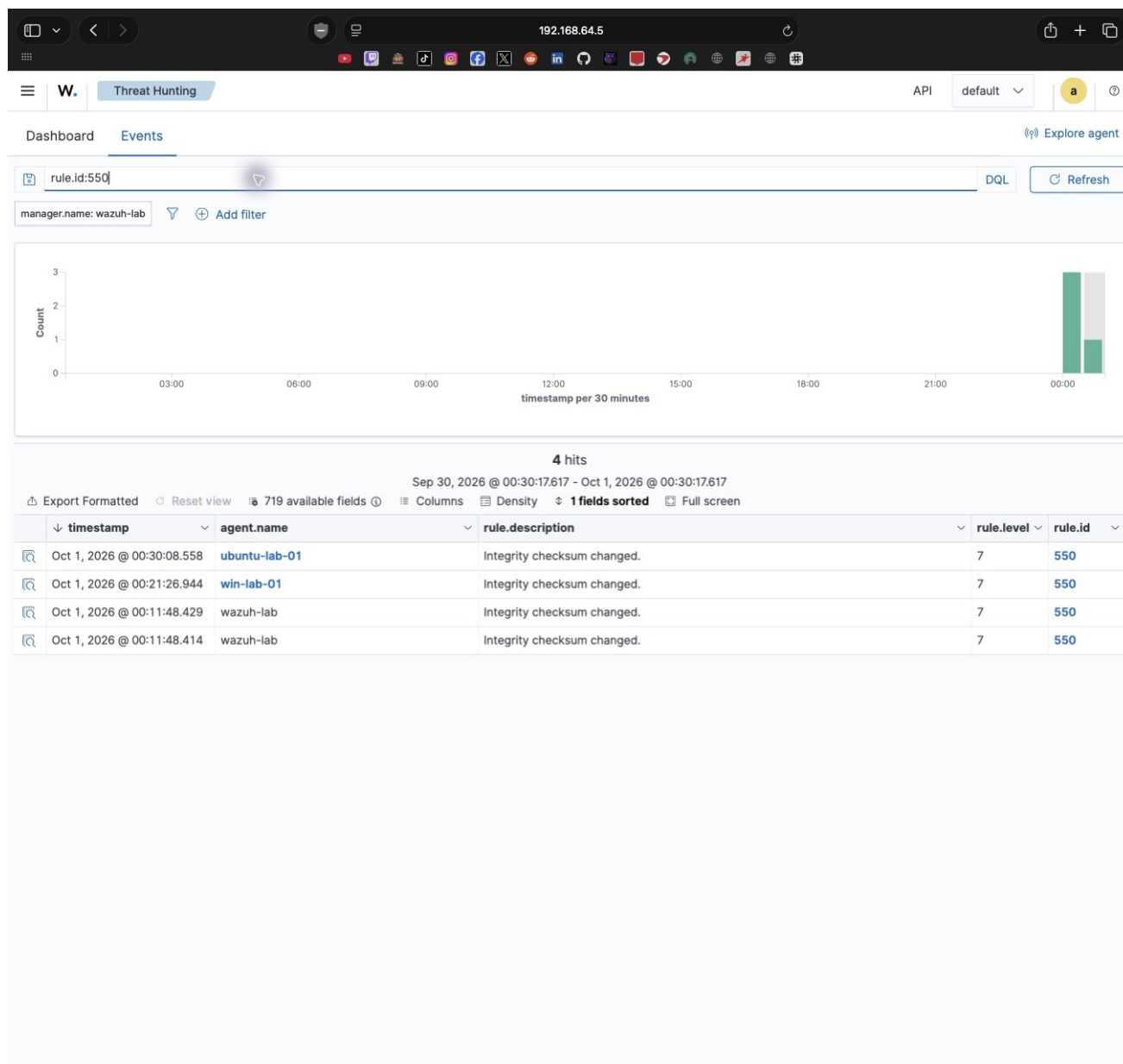
WZ04 The controlled alert shows win-lab-01, SocLabDoesNotExist, event 4625, and record 28868. Full fields remain available in the JSON evidence.

Observation: Windows logged the failed authentication and Wazuh retained the relevant fields.

Finding: Windows Security event collection and this authentication rule worked for the controlled test.

File Integrity Monitoring Results

Harmless baseline files were monitored in real time on Windows and Ubuntu. Their initial scans finished before the files were changed. Wazuh detected the modifications with rule 550 at level 7.

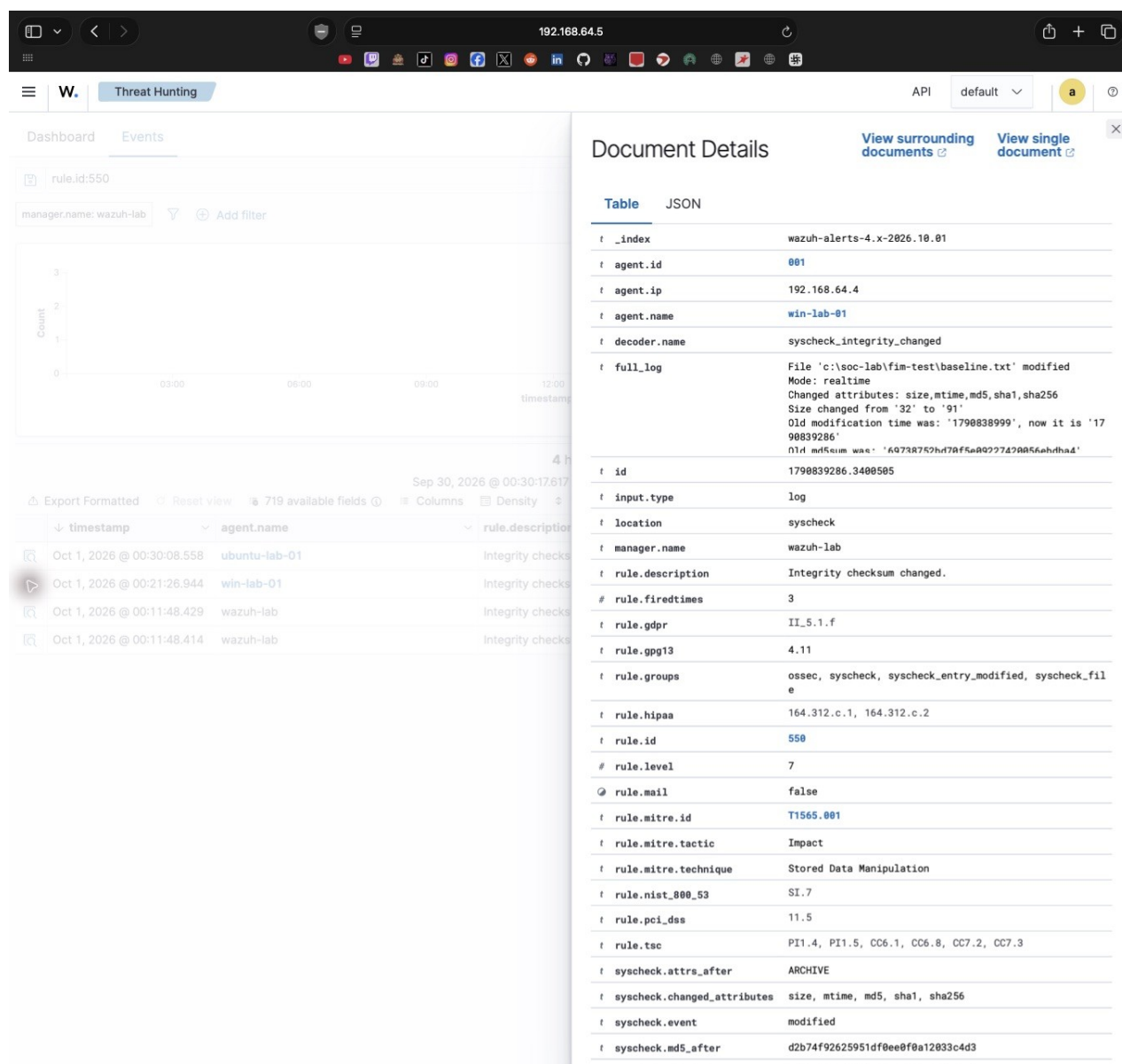


WZ05 Real file-change alerts from Ubuntu, Windows, and the server. The two earlier server alerts resulted from one overwrite passing through an empty-file state.

Windows baseline: 32 to 91 bytes. Ubuntu baseline: 30 to 87 bytes. Matching before-and-after hashes are preserved alongside the source change records.

Windows File Change Evidence

Windows monitored C:\SOC-Lab\FIM-Test in real time. The test appended harmless text to baseline.txt. Wazuh reported the same path, modified event, size change, and checksum changes.



WZ05 Windows file-change details show agent 001, the exact path, realtime mode, rule 550, and level 7.

Change recorded at 07:21:26.769 UTC; alert at 07:21:26.944 UTC. The observed delay was about 0.175 seconds for this test, not a general performance guarantee.

The file's SHA256 changed from 0fa8533c...24d83d55 to 41c8952d...9d5fd621. The complete hashes in WZ05-windows-change.json match the corresponding Wazuh fields.

Ubuntu Validation and Packet Correlation

Ubuntu file integrity test

Ubuntu monitors /opt/soc-lab/fim-test in real time. Its baseline scan ended at 07:29:22 UTC. The file changed at 07:30:08.524 UTC, and agent 002 produced rule 550 at 07:30:08.558 UTC. Size changed from 30 to 87 bytes. The measured delay was about 0.034 seconds.

A second append at 07:32:21.352 UTC changed the file from 87 to 144 bytes. The corresponding alert was generated at 07:32:21.385 UTC. This repeat occurred during the packet capture.

Network evidence

A 25-second capture on the server interface enp0s1 recorded 1,034 packets matching TCP port 1514 and the two endpoint addresses. The saved PCAP uses a 96-byte snapshot length; it preserves packet headers and only the start of each packet.

Source	Observed fact	Limit
Packet capture	192.168.64.3 and .4 communicate with .5 on TCP 1514	Headers establish transport; they do not identify an event payload.
Ubuntu source action	Second file append at 07:32:21.352 UTC	A planned action, not evidence of malicious editing.
Wazuh alert	Agent 002 file modification at 07:32:21.385 UTC	Temporal correlation does not map one alert to one encrypted packet.

Example captured packet at 07:32:10.301420 UTC: 192.168.64.3:35547 → 192.168.64.5:1514, TCP payload length 382 bytes. The server acknowledged it at 07:32:10.301492 UTC.

The Windows agent log confirms AES transport encryption. Packet timing and endpoint addresses complement the host evidence, but the capture does not prove the plaintext content of an event. Ubuntu cannot automatically observe all Windows traffic on Shared NAT.

Compatibility limit

The Ubuntu 26.04 agent installed and passed connectivity and FIM tests. Its bundled CIS Ubuntu 22.04 assessment policy was skipped because the OS version check failed. This lab does not claim Ubuntu 26.04 CIS assessment coverage.

Progress and Operating Notes

Phase	Status	Verified evidence
Baseline and recovery preparation	Complete	VM inventory and pre-agent snapshots
Central deployment	Complete	WZ02 dashboard and service checks
Endpoint enrollment	Complete	WZ03 two Active agents
Authentication investigation	Complete	WZ04 source event and alert
File integrity tests	Complete	WZ05 Windows and Ubuntu alerts
Packet correlation	Complete	WZ06 PCAP and repeat Ubuntu change
Analyst investigation	Complete	SOC2 AUTH 001 timeline and disposition

Troubleshooting completed

The cloud-image boot initially waited for networking. An explicit DHCP network configuration and VirtIO cloud-init seed resolved it. The central services then installed successfully.

Windows guest management was unavailable until its QEMU Guest Agent service ran. Its Wazuh agent initially showed Pending and later became Active. Source-event matching confirmed collection after enrollment.

The first Ubuntu package installation began before its transferred file finished copying and reported a truncated archive. Installation was repeated after the transfer completed; the package and service then verified successfully.

An initial dashboard query contained a nonbreaking space that Wazuh rejected. Searching rule.id:60 122 and rule.id:550 produced the expected records.

How to resume the lab

Start SOC Homelab 2 - Wazuh, then the endpoint VMs. Open <https://192.168.64.5> in the browser already trusted for this lab and use the existing private dashboard credentials. The login prompt inside the server VM is the Linux console; it is separate from the dashboard login.

Keep the central VM's disk, cloud-init seed, and UTM configuration together. Credentials and agent registration keys are private and are excluded from the evidence package. The monitoring folders remain enabled so you can repeat harmless tests.

Project Summary and References

This lab extends the first SOC Homelab with a working Wazuh server and monitored Windows and Ubuntu endpoints. A controlled Windows failed authentication was traced from its local Security log to Wazuh rule 60 122. Real-time file modifications on both endpoints triggered rule 550. Agent transport was captured and compared with a repeated Ubuntu file-change test.

The investigation closed as expected lab activity because the source event and alert matched the planned test. The project demonstrates endpoint enrollment, event validation, file integrity monitoring, evidence preservation, and contextual alert triage.

Evidence files

The companion evidence folder contains original screenshots, source JSON records, Wazuh alert exports, central service status, and the agent transport PCAP and text summary. Screenshot IDs correspond to the captions in this report. The full hashes and timestamps are retained in those files.

Limits and next steps

Only the exercised sources are validated. This project does not establish full process, DNS, network, malware, vulnerability, or CIS coverage. Ubuntu 26.04 assessment policy support remains a gap. A supported Ubuntu endpoint, selected additional telemetry, and successful-login correlation are useful future improvements.

Reference material

[1] Wazuh Quickstart

<https://documentation.wazuh.com/current/quickstart.html>

[2] Wazuh components

<https://documentation.wazuh.com/current/getting-started/components/index.html>

[3] Windows agent installation

<https://documentation.wazuh.com/current/installation-guide/wazuh-agent/wazuh-agent-package-windows.html>

[4] Linux agent installation

<https://documentation.wazuh.com/current/installation-guide/wazuh-agent/wazuh-agent-package-linux.html>

[5] File integrity monitoring settings

<https://documentation.wazuh.com/current/user-manual/capabilities/file-integrity/basic-settings.html>

[6] Official Ubuntu ARM64 cloud image

<https://cloud-images.ubuntu.com/noble/current/noble-server-cloudimg-arm64.img>

Prior project reference: SOC Homelab / Wireshark Network Traffic Investigation. Historical findings were used as background, and the Wazuh results above come from this lab's tests.